



NORTH SYDNEY BOYS' HIGH SCHOOL
2008 HSC Course Assessment Task 2

MATHEMATICS

General instructions

- Working time – 60 minutes.
- Write in the booklet provided.
- Each new question is to be started on a new booklet.
- Write using blue or black pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets within this paper and hand to examination supervisors.

Class teacher (please ✓)

- ☐ Mr Fletcher
- ☐ Mr Lam
- ☐ Mr Lowe
- ☐ Mr Ireland
- ☐ Mr Trenwith
- ☐ Mr Rezcallah
- ☐ Mr Weiss

STUDENT NUMBER:

Marker's use only.

QUESTION	MARKS
1	/12
2	/12
3	/11
4	/13
5	/6
Total	/54
Total (%)	/100

Question 1 (12 Marks)		Commence a new booklet.	Marks
(a)	i.	Find $\int 6x^7 dx$.	2
	ii.	Evaluate $\int_0^1 5x^4 + 3x + 1 dx$.	2
(b)		Find $\int \frac{x^3 + x^2}{2x} dx$.	2
(c)	i.	Differentiate $y = (2x^5 - 1)^3$.	2
	ii.	Hence or otherwise, find $\int 10x^4 (2x^5 - 1)^2 dx$.	2
(d)		Find the <i>exact</i> value of k if $\int_2^k 3x^2 dx = 50$.	2

Question 2 (12 Marks)		Commence a new booklet.	
(a)	i.	Evaluate $\int_{-a}^a x^5 dx$.	2
	ii.	Find the area bounded by the curve $y = x^5$ between the lines $x = -a$, $x = a$ and the x axis.	2
(b)		By sketching the curve $y = \sqrt{9 - x^2}$, hence or otherwise evaluate $\int_0^3 \sqrt{9 - x^2} dx$ as an exact value.	3
(c)		Given $s''(t) = 2t^2$, $s'(2) = 1$ and $s(1) = 2$, find:	
	i.	$s'(t)$	2
	ii.	$s(3)$	3

Question 3 (11 Marks) Commence a **new** booklet. **Marks**

- (a) Find the area of the region between the curves **4**

$$\begin{cases} y = x^2 \\ y = 3x + 4 \end{cases}$$

- (b) Find the volume of solid of revolution when the region bounded by the curve **3**
 $y = x^2 + 1$, the y axis, the lines $y = 2$ & $y = 5$, rotated about the y axis.

- (c) Find an approximation to $\int_0^1 2^x dx$ by using Simpson's Rule with 5 function **4**
values.

Question 4 (13 Marks) Commence a **new** booklet.

- (a) The probability that a particular man lives to the age of 75 is $\frac{4}{5}$ and the probability that his wife will live to 75 is $\frac{6}{7}$. By drawing a tree diagram or otherwise, find:
- i. Only the man will live to 75. **1**
 - ii. Both will live to 75. **2**
 - iii. At least one of them will live to 75. **2**

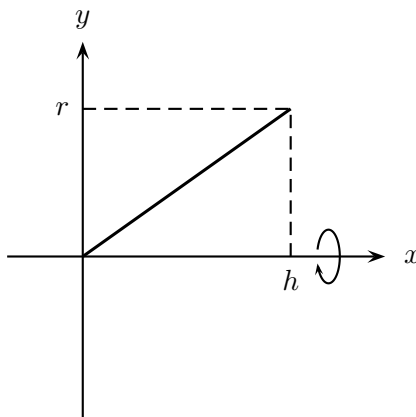
- (b) In a bag with 20 marbles, seven are red, nine are gold & four are blue. One marble is taken from the bag and not replaced, then a second is taken out.

Find the probability of choosing:

- i. Two red marbles. **2**
 - ii. Marbles of a different colour **2**
- (c) If an integer x between 1 and 100 (inclusive) is chosen at random, find the probability of the number being:
- i. Less than 50 or a multiple of 5. **2**
 - ii. Being a multiple of 9 but not a multiple of 12. **2**

Question 5 (6 Marks)Commence a **new** booklet.**Marks**

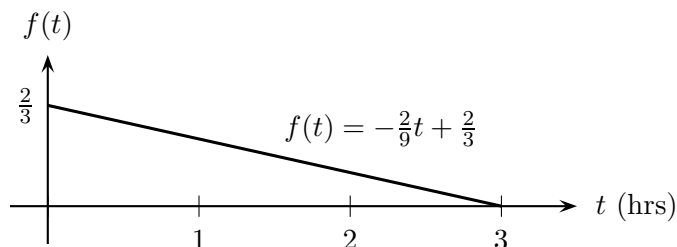
- (a) i. Find the equation of a line passing through the points $(0,0)$ and (h,r) . **1**
- ii. Using the diagram below & by rotating the given line about the x axis, derive the formula for the volume of a cone with radius r and height h . **2**



- (b) A certain telecommunications company offers “untimed” international calls but disconnects the call after 3 hours.

The probability of a call duration between $a < t < b$ hours using the function is

$$P(a < t < b) = \int_a^b f(t) dt$$



Using this, find the probability of a call lasting

- i. Between 60 & 90 minutes. **2**
- ii. *Exactly* 1.5 hours. **1**

End of paper.

Solutions

Question 1

(a) i. (2 marks)

$$\int 6x^7 dx = \frac{3}{4}x^8 + C$$

ii. (2 marks)

$$\begin{aligned} & \int_0^1 5x^4 + 3x + 1 dx \\ &= \left[x^5 + \frac{3}{2}x^2 + x \right]_0^1 \\ &= 1 + \frac{3}{2} + 1 = \frac{7}{2} \end{aligned}$$

(b) (2 marks)

$$\begin{aligned} \int \frac{x^3 + x^2}{2x} dx &= \frac{1}{2} \int \frac{x^3}{x} + \frac{x^2}{x} dx \\ &= \frac{1}{2} \int x^2 + x dx \\ &= \frac{1}{2} \left(\frac{1}{3}x^3 + \frac{1}{2}x^2 \right) + C \\ &= \frac{1}{6}x^3 + \frac{1}{4}x^2 + C \end{aligned}$$

(c) i. (2 marks)

$$\begin{aligned} y &= (2x^5 - 1)^3 \\ \left. \begin{aligned} y(u) &= u^3 & u(x) &= 2x^5 - 1 \\ y'(u) &= 3u^2 & u'(x) &= 10x^4 \end{aligned} \right\} \\ y'(x) &= y'(u) \times u'(x) \\ &= 3u^2 \times 10x^4 \\ &= 30x^4 (2x^5 - 1)^2 \end{aligned}$$

ii. (2 marks)

$$\begin{aligned} & \int 10x^4 (2x^5 - 1)^2 dx \\ &= \frac{1}{3} \int 30x^4 (2x^5 - 1)^2 dx \\ &= \frac{1}{3} (2x^5 - 1)^3 + C \end{aligned}$$

iii. (2 marks)

$$\begin{aligned} \int_2^k 3x^2 &= 50 \\ \left[x^3 \right]_2^k &= 50 \\ k^3 - 2^3 &= 50 \\ k^3 &= 58 \\ k &= 58^{1/3} \end{aligned}$$

Question 2

(a) i. (2 marks)

$$\begin{aligned} \int_{-a}^a x^5 dx &= \left[\frac{1}{6}x^6 \right]_{-a}^a \\ &= \frac{1}{6} (\cancel{a^6} - (\cancel{-a})^6) \\ &= 0 \end{aligned}$$

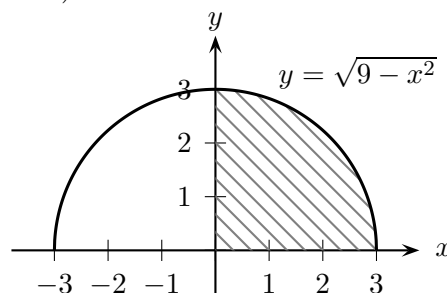
ii. (2 marks)

Since the integral is 0, then the area from $x = -a$ to $x = 0$ must be equal to the area from $x = 0$ to $x = a$.

$$\begin{aligned} A &= 2 \int_0^a x^5 dx \\ &= 2 \left[\frac{1}{6}x^6 \right]_0^a \\ &= \frac{1}{3}a^6 \end{aligned}$$

Both marks to be awarded iff explanation is acceptable.

(b) (3 marks)



The integral $\int_0^3 \sqrt{9 - x^2} dx$ is the same as the area of the quarter circle.

$$\begin{aligned} \int_0^3 \sqrt{9 - x^2} dx &= \frac{1}{4}\pi \times 3^2 \\ &= \frac{9}{4}\pi \end{aligned}$$

(c) i. (2 marks)

$$\begin{aligned}
 s'(t) &= \int s''(t) dt \\
 &= \int 2t^2 dt \\
 &= \frac{2}{3}t^3 + C_1
 \end{aligned}$$

Using $s'(2) = 1$

$$\begin{aligned}
 1 &= \frac{2}{3} \times 2^3 + C_1 \\
 &= \frac{16}{3} + C_1 \\
 C_1 &= 1 - \frac{16}{3} = -\frac{13}{3} \\
 \therefore s'(t) &= \frac{2}{3}t^3 - \frac{13}{3}
 \end{aligned}$$

ii. (3 marks)

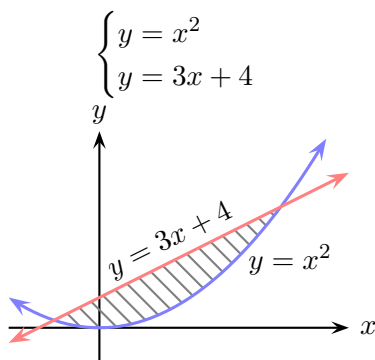
$$\begin{aligned}
 s(t) &= \int s'(t) dt \\
 &= \int \left(\frac{2}{3}t^3 - \frac{13}{3} \right) dt \\
 &= \frac{2}{3} \cdot \frac{1}{4}t^4 - \frac{13}{3}t + C_2 \\
 &= \frac{1}{6}t^4 - \frac{13}{3}t + C_2
 \end{aligned}$$

Using $s(1) = 2$,

$$\begin{aligned}
 2 &= \frac{1}{6} - \frac{13}{3} + C_2 \\
 C_2 &= 2 - \frac{1}{6} + \frac{13}{3} \\
 &= \frac{37}{6} \\
 \therefore s(t) &= \frac{1}{6}t^4 - \frac{13}{3}t + \frac{37}{6} \\
 \therefore s(3) &= \frac{20}{3}
 \end{aligned}$$

Question 3

(a) (4 marks)



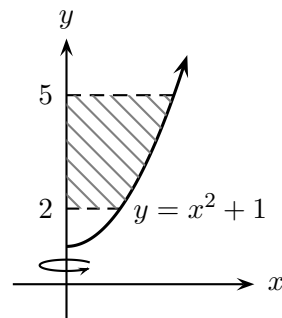
Find the points of intersection between the two curves by equating,

$$\begin{aligned}
 x^2 &= 3x + 4 \\
 x^2 - 3x - 4 &= 0 \\
 (x - 4)(x + 1) &= 0 \\
 \therefore x &= 4, -1
 \end{aligned}$$

The area is thus

$$\begin{aligned}
 A &= \left| \int_{-1}^4 (3x + 4) - x^2 dx \right| \\
 &= \left| \int_{-1}^4 x^2 - 3x - 4 dx \right| \\
 &= \left| \left[\frac{1}{3}x^3 - \frac{3}{2}x^2 - 4x \right]_{-1}^4 \right| \\
 &= \frac{125}{6}
 \end{aligned}$$

(b) (3 marks)

Changing the subject to x^2 :

$$\begin{aligned}
 y &= x^2 + 1 \\
 x^2 &= y - 1
 \end{aligned}$$

Integrating,

$$\begin{aligned}
 V &= \pi \int_2^5 x^2 dy \\
 &= \pi \int_2^5 (y - 1) dy \\
 &= \pi \left[\frac{1}{2}y^2 - y \right]_2^5 \\
 &= \pi \left(\frac{1}{2}(5^2 - 2^2) - (5 - 2) \right) \\
 &= \pi \left(\frac{21}{2} - 3 \right) \\
 &= \frac{15\pi}{2}
 \end{aligned}$$

(c) (4 marks)

[2] if the entire table is correct

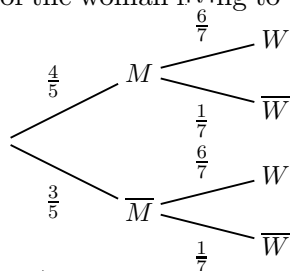
x	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1
2^x	1	$2^{\frac{1}{4}}$	$2^{\frac{1}{2}}$	$2^{\frac{3}{4}}$	2

Using Simpson's Rule with the 5 function values from the table,*

$$\begin{aligned}
 A &\approx \frac{h}{3} (y_1 + 4y_{\text{even}} + 2y_{\text{odd}} + y_{\ell}) \\
 &= \frac{\frac{1}{4}}{3} \left(1 + 4 \left(2^{1/4} + 2^{3/4} \right) + 2 \times 2^{1/2} + 2 \right) \\
 &= 1.4427 \text{ (4 d.p.)}
 \end{aligned}$$

Question 4

- (a) Draw out the probability tree, letting the event of the man living to 75+ be M and the event of the woman living to 75+ be W .



i. (1 mark)

$$P(M) = P(M) \times P(\overline{W}) = \frac{4}{5} \times \frac{1}{7} = \frac{4}{35}$$

ii. (2 marks)

$$\begin{aligned}
 P(MW) &= P(M) \times P(W) \\
 &= \frac{4}{5} \times \frac{6}{7} = \frac{24}{35}
 \end{aligned}$$

The first mark is for identification of a pair of independent events.

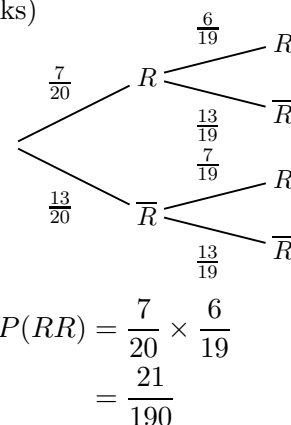
iii. (2 marks)

$$\begin{aligned}
 &P(\text{at least } M \text{ or } W) \\
 &= 1 - P(\text{neither}) \\
 &= 1 - (P(\overline{M})P(\overline{W})) \\
 &= 1 - \left(\frac{3}{5} \times \frac{1}{7} \right) = \frac{32}{35}
 \end{aligned}$$

The first mark is for identifying the complement.

*Students may use a 4 d.p. approximation of 2^x where 2^x is irrational.

(b) i. (2 marks)



ii. (2 marks)

$$\begin{aligned}
 &P(\text{different}) \\
 &= 1 - P(\text{same}) \\
 &= 1 - (P(RR) + P(GG) + P(BB)) \\
 &= 1 - \left(\frac{21}{190} + \left(\frac{9}{20} \times \frac{8}{19} \right) + \left(\frac{4}{20} \times \frac{3}{19} \right) \right) \\
 &= \frac{127}{190}
 \end{aligned}$$

- (c) A Venn diagram would assist in the calculations. Not drawn here for brevity.

i. (2 marks)

There are 11 multiples of 5 from 50 to 100 inclusive, and 49 numbers less than 50 (1 to 49). Hence

$$P(x < 50 \cup 5|x) = \frac{49 + 11}{100} = \frac{3}{5}$$

A Venn diagram would assist in the calculations.

ii. (2 marks)

Common multiples of 9 and 12 are 36 & 72. Since there are 11 multiples of 9 between 1 & 100, then there are 9 of them which are not also multiples of 12. Hence

$$P(x|9 \cap 12 \not|x) = \frac{9}{100}$$

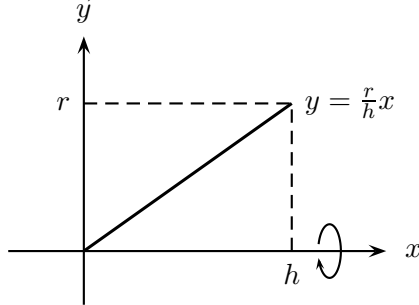
Question 5

(a) i. (1 mark)

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{r - 0}{h - 0} = \frac{r}{h}$$

$$\therefore y = \frac{r}{h}x$$

ii. (2 marks)



$$V = \pi \int_0^h y^2 dx$$

$$= \pi \int_0^h \frac{r^2}{h^2} x^2 dx$$

$$= \frac{\pi r^2}{h^2} \left[\frac{1}{3} x^3 \right]_0^h$$

$$= \frac{\pi r^2}{h^2} \times \frac{1}{3} h^3$$

$$= \frac{1}{3} \pi r^2 h$$

(b) i. (2 marks)

$$P(1 < t < 1.5)$$

$$= \int_1^{3/2} -\frac{2}{9}t + \frac{2}{3} dt$$

$$= \left[-\frac{1}{9}t^2 + \frac{2}{3}t \right]_1^{3/2}$$

$$= \left(-\frac{1}{9} \left(\frac{9}{4} - 1 \right) + \frac{2}{3} \left(\frac{3}{2} - 1 \right) \right)$$

$$= \left(-\frac{1}{9} \left(\frac{5}{4} \right) + \frac{2}{3} \times \frac{1}{2} \right)$$

$$= \frac{7}{36}$$

Both marks to be awarded if student is capable of finding the function values at $t = 1$ and $t = \frac{3}{2}$, then using the area of a trapezium to find the proper area.

ii. (1 mark)

$$P(t = 1.5) = P(1.5 < t < 1.5)$$

$$= \int_{1.5}^{1.5} f(t) dt$$

$$= 0$$

Highlights the theoretical impossibility of ANY call lasting exactly 1.5 hours under a continuous probability distribution.